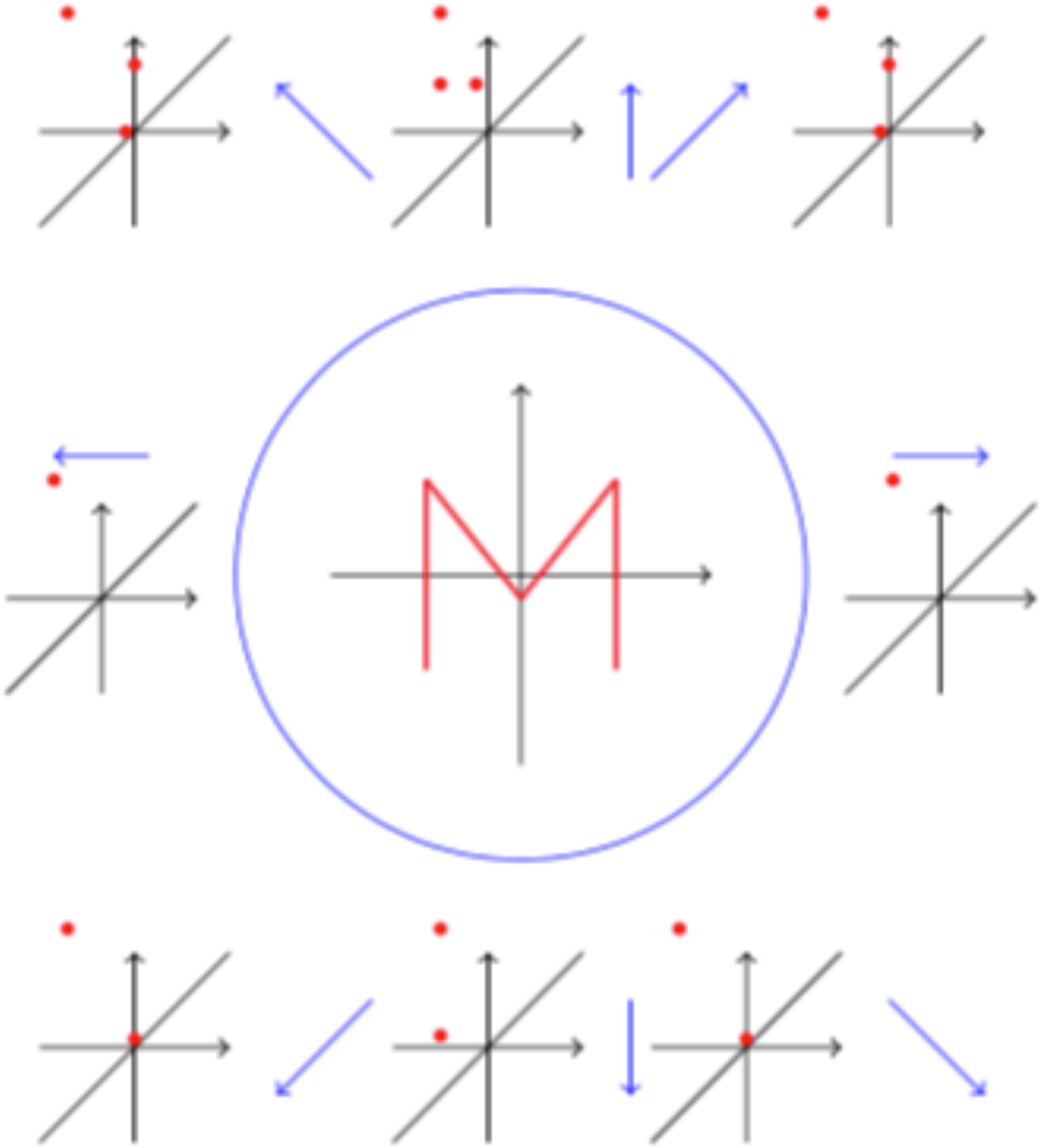

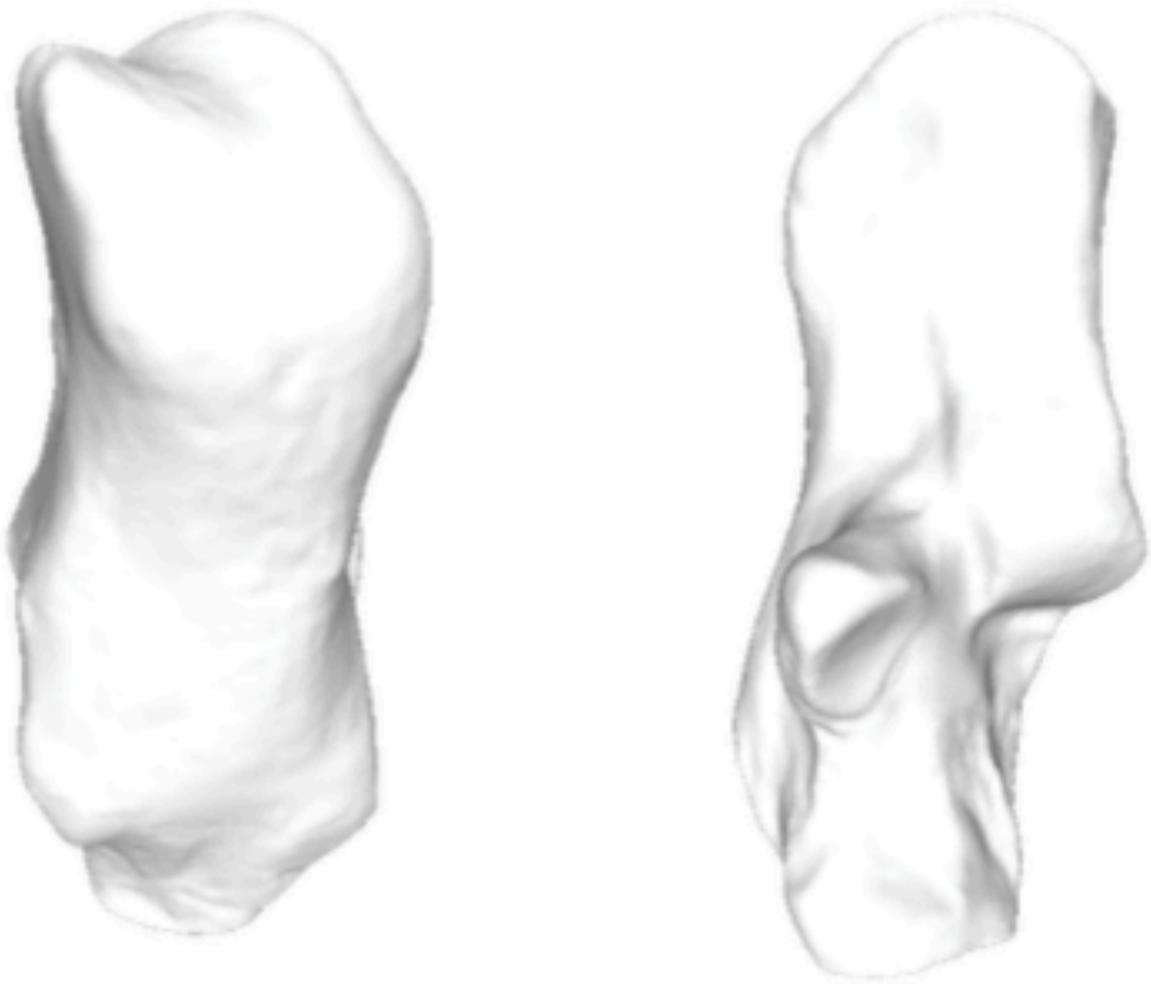

Figure 13: Images of a calcaneus from two different angles.

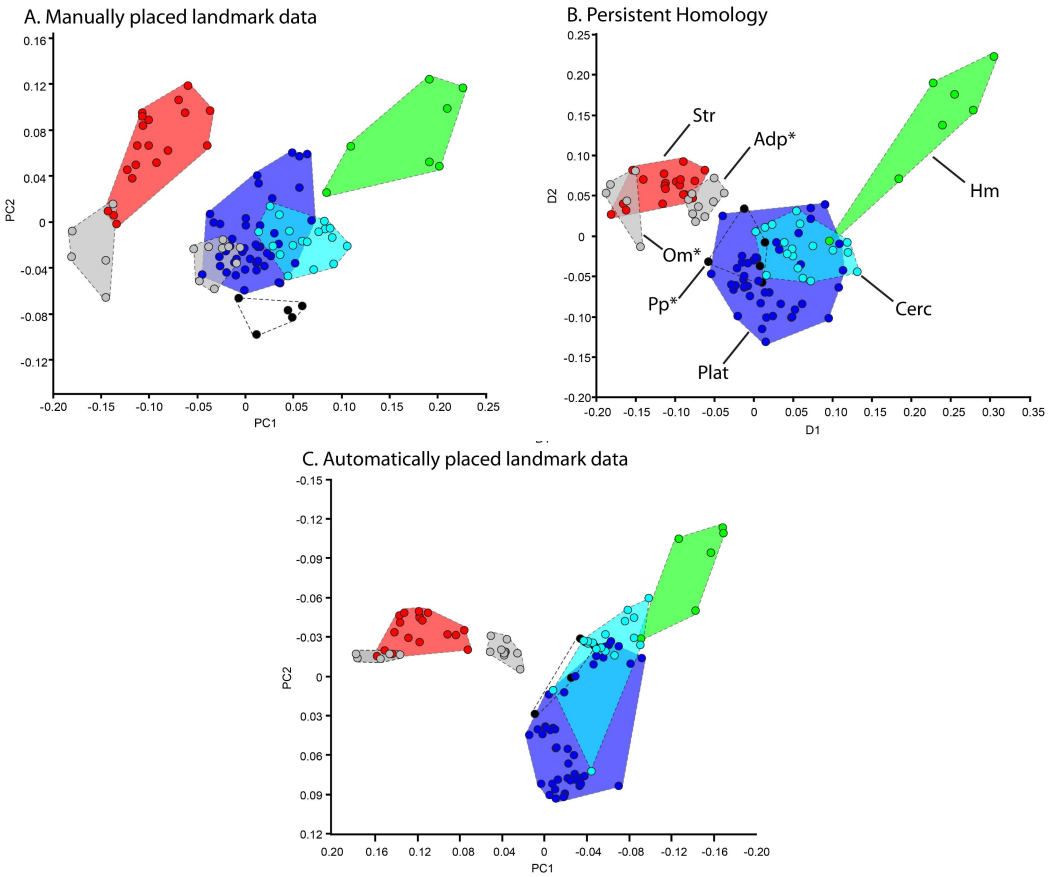

Figure 14: Phenetic clustering of phylogenetic groups of primate calcanei ($n = 106$). 67 genera are represented. See appendix for specimen information. Asterisks indicate groups of extinct taxa. Abbreviations: Str, Strepsirrhines; Plat, platyrrhines; Cerc, Cercopithecoids; Om, Omomyiforms; Adp, Adapiforms; Pp, parapithecids; Hmn, Hominoids. Note that more primitive prosimian taxa cluster separately from simians (Om, Adp, Str.). Also note that monkeys (Plat, Cerc, Pp) cluster mainly separately from apes (Hmn). This is virtually the same type of clustering observed with more labor intensive (and fundamentally subjective) manually collected landmarks. The position of adapiforms between strepsirrhines and platyrrhines is consistent with previous descriptions of the phenetic affinities of this group. (a) is using the manual alignment method, (b) is using the PHT method, (c) is the automated landmark based method.

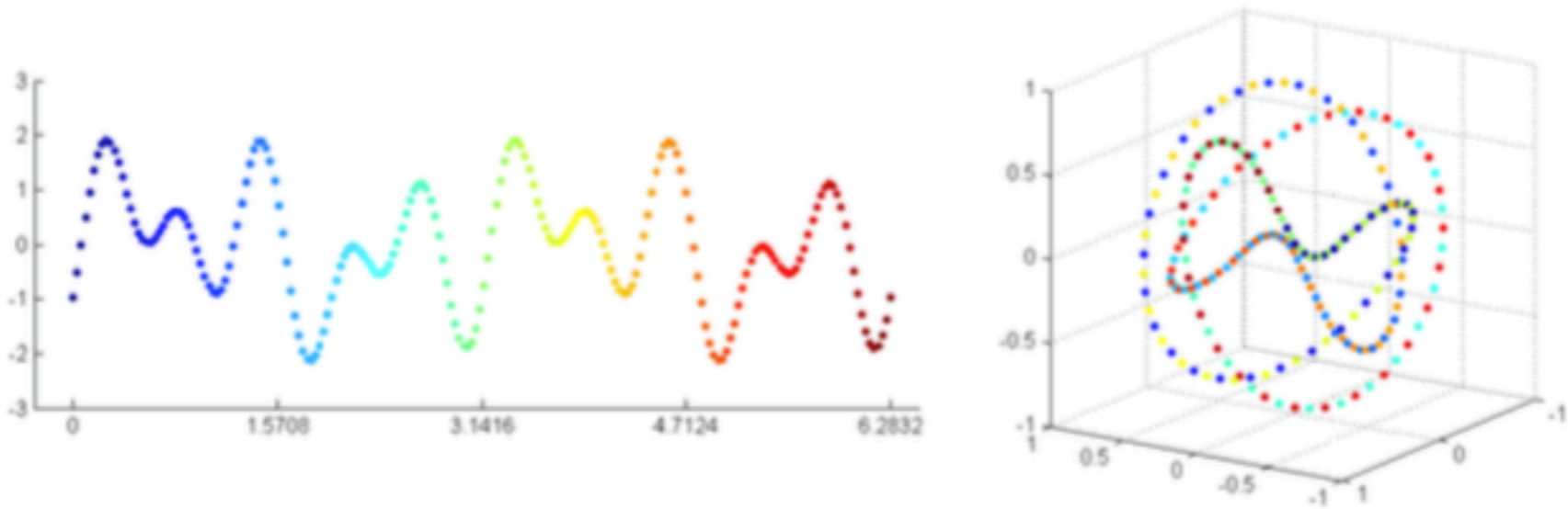

FIGURE 1. From a periodic function to its sliding window point cloud. **Left:** A periodic function f. **Right:** Multidimensional scaling into $\mathbb{R}^3$ for $SW_{20,\tau}f$. For each t, we use the same color for $f(t)$ and $SW_{20,\tau}f(t)$. Please refer to an electronic version for colors.

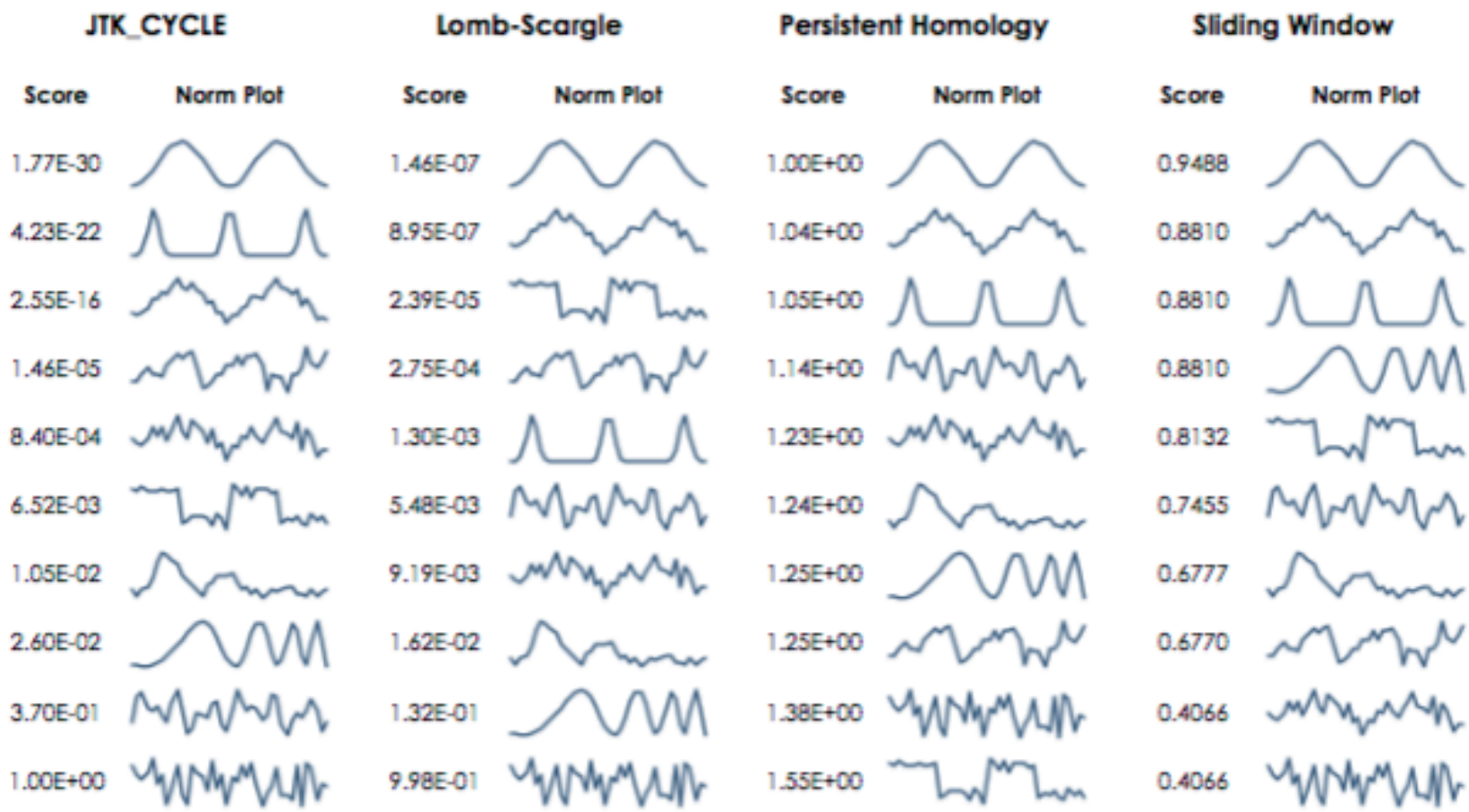

FIGURE 4. Ranking of signals by periodicity. For each algorithm we provide the score and a normalized plot of the signal. The ranking goes from top (highest score) to bottom (lowest score).